

Chapter 7

Pragmatics and Discourse Analysis

Overview:

1. Monologue and Dialogue
2. Reference Resolution
3. Coherence and Cohesion
4. Discourse Structure

7.1 Pragmatics in NLP

- Pragmatics is a subfield of linguistics.
- It studies how context influences the way language is interpreted and understood.
- It goes beyond the literal meaning of words to consider how speakers use language in real situations and how listeners interpret utterances based on various factors, including the situation, the relationship between speakers, and shared knowledge.

- **Pragmatics** is the branch of linguistics and NLP that studies **how people understand meaning using context**, not just grammar or dictionary meaning.
- Pragmatics answers the question:
“What does the speaker really mean?”

- Consider the following dialogue between two people:
 - **Person A:** "Can you pass the salt?"
 - **Person B:** "Sure."
- In this example:
 - **Literal Meaning:** Person A is asking whether Person B has the ability to pass the salt.
 - **Pragmatic Meaning:** Person A is actually requesting that Person B pass the salt to them. Person B understands this and responds accordingly by agreeing to pass the salt.

- **Another Example**
 - **Person A:** "It's cold in here."
 - **Person B:** (closes the window)
- In this case:
 - **Literal Meaning:** Person A is making a statement about the temperature.
 - **Pragmatic Meaning:** Person A is implying that they would like the window to be closed to warm up the room. Person B understands the implied request and acts on it.

- Sentence:
 - “It’s getting late.”
- Possible pragmatic meanings:
 - Let’s end the meeting
 - I want to go home
 - Stop talking
- Same sentence → different meanings → **context decides**

Why Pragmatics is Important in NLP

- Without pragmatics:
 - Chatbots reply wrongly
 - Machine translation sounds rude
 - Sentiment analysis fails

Key Concepts in Pragmatics

1. **Deixis:** Words or phrases (e.g., "this," "that," "here," "there") that require contextual information to be understood.
 - **Example:** "I'll meet you here tomorrow."
 - "here" and "tomorrow" depend on the context of the conversation.
2. **Implicature:** The implied meaning that goes beyond the literal meaning of the words.
 - **Example:** "Can you pass the salt?"
 - This is typically understood as a request rather than a question about one's ability.
3. **Speech Acts:** Actions performed via utterances, such as requests, promises, apologies, etc.
 - **Example:** "I apologize for being late."
 - This is an act of apology.
4. **Presupposition:** Assumptions the speaker makes about what the listener already knows.
 - **Example:** "John's brother is coming to the party."
 - Presupposes that John has a brother.

7.2 Discourse Integration in NLP

- **Discourse** in NLP refers to any connected piece of speech or writing that extends beyond a single sentence.
 - It involves the study and analysis of language use in texts and conversations to understand how sentences and utterances are related and how they contribute to the overall meaning.
- **Discourse Integration** refers to the process of combining information from multiple sentences or utterances to form a coherent and cohesive understanding of a text or conversation.
 - It involves understanding the relationships and dependencies between sentences and ensuring that the discourse is logically and contextually consistent.

Discourse Integration includes:

1. **Coherence:** The logical connections and consistency that make a text understandable.

Example: "John went to the store. He bought some milk."

The second sentence logically follows from the first.

2. **Cohesion:** The grammatical and lexical linking within text or speech that holds a discourse together.

Example: "The cat chased the mouse. It caught it."

Pronouns "it" refer to "The cat" and "the mouse," linking the sentences.

Discourse Structure:

- Discourse structure refers to **the way in which a coherent sequence of sentences or utterances is organized** and connected to form a meaningful text or conversation.
- It **encompasses various linguistic and cognitive elements** that help convey information effectively and maintain coherence.
- **Understanding discourse structure is essential** for various NLP applications such as text summarization, machine translation, dialogue systems, and more.
- Types of discourse structure:
 1. Narrative
 2. Expository
 3. Argumentative
 4. Descriptive

Types of Discourse Structures:

1. Narrative Structure:

- Organizes events or actions in a sequence, often with a clear beginning, middle, and end.
- Common in storytelling and historical accounts.
- **Example:** "First, he woke up. Then, he had breakfast. Finally, he left for work."

2. Expository Structure:

- Presents information or explains a concept in a logical order.
- Common in academic writing, technical manuals, and reports.
- **Example:** "Photosynthesis is a process used by plants. It involves the conversion of light energy into chemical energy."

3. **Argumentative Structure:**

- Organizes text around arguments, counter-arguments, and evidence.
- Common in persuasive essays and debates.
- Example: "Smoking should be banned in public places. Studies show it harms non-smokers through second-hand smoke."

4. **Descriptive Structure:**

- Provides detailed descriptions of objects, people, or settings.
- Common in literature and observational reports.
- Example: "The room was dimly lit, with shadows dancing on the walls. A musty smell filled the air."

7.3 Monologue in NLP:

- A **monologue involves a single speaker** or writer expressing their thoughts without any interaction or exchange with others.
- Examples include speeches, lectures, essays, or personal narratives.
- Key tasks in NLP related to monologues include:
 - **Text Summarization**: Creating concise summaries of long monologues.
 - **Sentiment Analysis**: Determining the sentiment or emotional tone of a monologue.
 - **Topic Modeling**: Identifying the main topics or themes within a monologue.
 - **Named Entity Recognition (NER)**: Extracting names of people, places, organizations, and other entities mentioned in the monologue.
 - **Speech-to-Text**: Converting spoken monologues into written text.

7.4 Dialogue in NLP:

- A dialogue involves an **interactive exchange between two or more participants**. Examples include conversations, interviews, and chat interactions.
- Key tasks in NLP related to dialogues include:
 - **Dialogue Systems**: Building chatbots and virtual assistants that can interact with users naturally.
 - **Dialogue Management**: Handling the flow of conversation, including turn-taking and context management.
 - **Intent Recognition**: Identifying the intent behind each participant's utterance in the dialogue.
 - **Response Generation**: Automatically generating appropriate responses in a dialogue.
 - **Speaker Identification**: Distinguishing between different speakers in a conversation.

7.5 Types of Reference:

In Natural Language Processing (NLP), reference involves identifying the relationships between linguistic elements and the entities or concepts they refer to.

1. Anaphoric Reference
2. Cataphoric
3. Exophoric
4. Endophoric
5. Homophoric
6. Coreference
7. Bridging

1. Anaphoric Reference

- Anaphoric reference occurs when **a word or phrase refers back to another word or phrase mentioned earlier in the text.** This is common with pronouns and other anaphoric expressions.

Example:

- "John saw a movie. He liked it."
 - "He" refers back to "John" (anaphor and antecedent).
 - "it" refers back to "a movie" (anaphor and antecedent).

2. Cataphoric Reference

- Cataphoric reference is the opposite of anaphoric reference.
- It occurs when **a word or phrase refers forward to another word or phrase that appears later in the text.**

Example:

- "Before he could finish, John left the room."
 - "he" refers forward to "John".

3. Exophoric Reference:

- Exophoric reference **involves referring to something outside the text**, often relying on the shared knowledge between the speaker and the listener.

Example:

- "Look at that!"
- "that" refers to something in the immediate physical environment or context outside the text.

4. Endophoric Reference:

- Endophoric reference refers to expressions that relate to other elements within the same text. This category includes both anaphoric and cataphoric references.

Example:

- "When she arrived, Mary noticed the lights were on."
- "She" refers to "Mary" (anaphoric), and "the lights" refers to something within the discourse context (endophoric).

5. Homophoric Reference:

- Homophoric reference involves a reference that can only be interpreted correctly with context-specific knowledge.
- It's a subset of exophoric reference where the context is general cultural or situational knowledge.

Example:

- "The Queen is visiting next week."
 - "The Queen" relies on the reader's or listener's knowledge of which queen is being referred to, depending on the context (e.g., the Queen of England).

6. Coreference:

- Coreference occurs when multiple expressions in the text refer to the same entity. It includes both anaphoric and cataphoric references but emphasizes the fact that different terms refer to the same thing.

Example:

- "Alice said she would come. She is very reliable."
 - "Alice" and "she" refer to the same entity, Alice.

7. Bridging Reference:

- Bridging reference involves referring to something that is not explicitly mentioned but is inferable from the context.
- **Example:** "I walked into the room. The windows were open."
 - "The windows" implies there are windows in the room mentioned earlier.

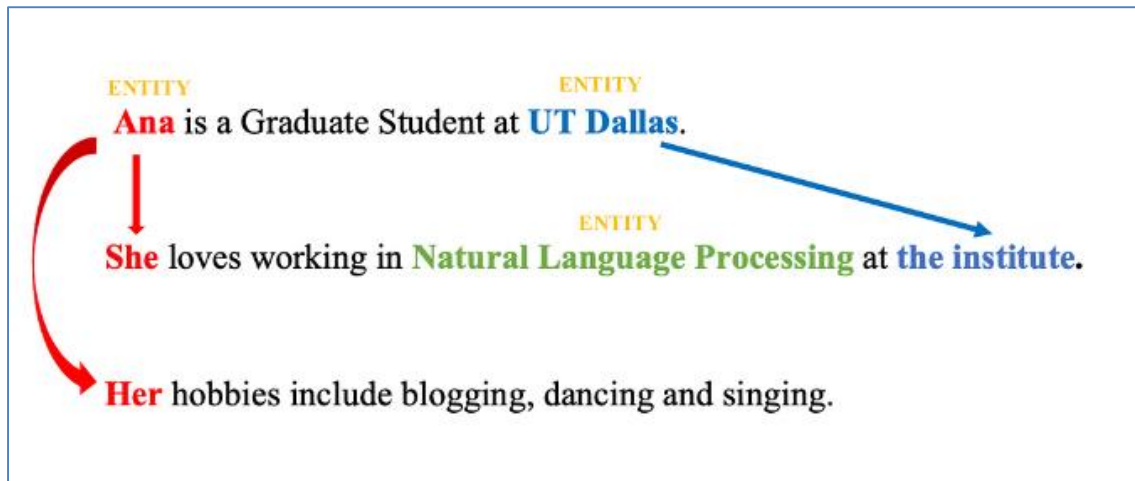
7. Reflexive Reference

- Reflexive reference occurs when a reflexive pronoun refers back to the subject of the clause.
- **Example:** "She prepared herself for the exam."
 - "Herself" refers back to "She."

7.6 Reference Resolution:

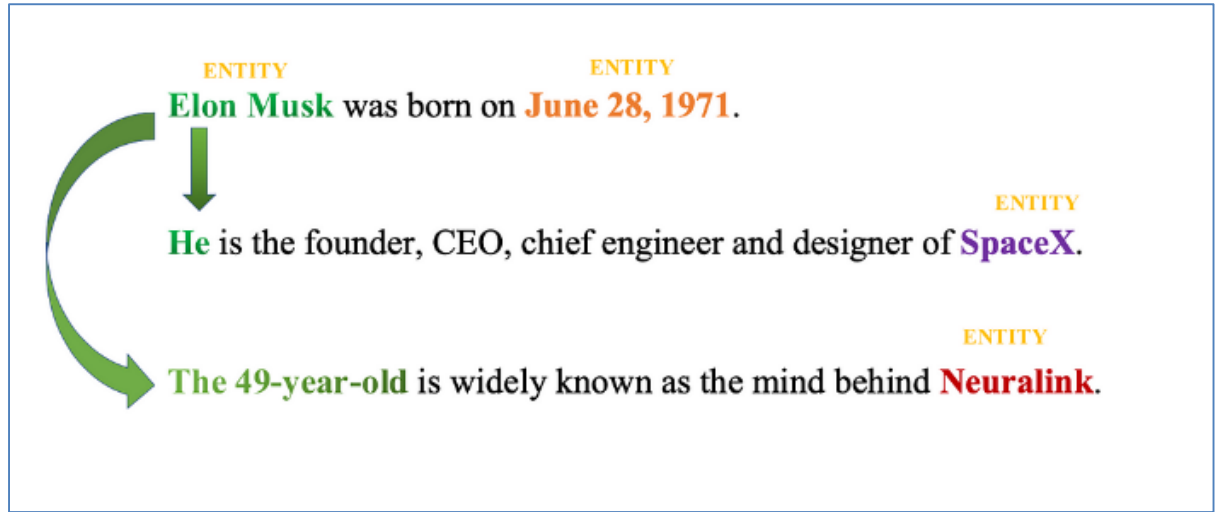
- Aka Coreference resolution
- involves identifying when different expressions in a text refer to the same entity.
- This can include pronouns, noun phrases, or other referring expressions.
- the process of resolving pronouns to identify which entities are they referring to.
- Proper reference resolution is essential for understanding the meaning of a text
- used in various NLP applications such as information extraction, machine translation, and text summarization.

Example 1:



- is the process of resolving pronouns to identify which entities are they referring to.
- The entities resolved may be a person, place, organization, or event.
- Referent is the object that is being referred to. For example, “Ana” is the referent in the above example.
- Referring expression are the mentions or linguistic expressions given in the discourse.
- Two or more referring expressions that refer to the same discourse entity are said to corefer.

Example 2.



- Referring Expressions: Elon Musk, He, The 49 year old
- Referent: Elon Musk
- Co-referring Expressions: {Elon Musk, He}, {Elon Musk, The 49 year old}

7.7 Challenges in Reference Resolution:

1. **Ambiguity:** Pronouns like "he," "she," "it," and "they" can be ambiguous if there are multiple possible antecedents.
2. **Long-Distance Dependencies:** The antecedent might be far from the referring expression, making it harder to resolve.
3. **Complex Structures:** Sentences with multiple clauses and nested structures can complicate reference resolution.
4. **Inference Requirement:** Bridging references often require inferencing and world knowledge to resolve correctly.

End of chapter 7